北京邮电大学 本科毕业设计（论文）任务书

Project Specification Form

Part 2 - Student

学院 **School**	International School	**专业** **Programme**	**Internet of Things Engineering**		
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论文题目 **Project Title**	AI Assisted Epilepsy detection using EEG signals				
论文概述 **Project outline** **Write about 500-800 words** **Please refer to Project Student Handbook section 3.2**	**Problem Analysis:** Epilepsy is a neurological disease affecting almost 50 million people worldwide. Recurrent seizures can cause injury, cognitive decline, and social stigma. **The core problem is**: how to detect seizures automatically and accurately from scalp EEG in real clinical settings, where signals are noisy, patients vary in age and recording conditions, and traditional manual review is time-consuming and subjective. This project aims to develop a Deep Learning Network (DLN) model to automatically classify EEG signals as epileptic or non-epileptic, thereby assisting neurologists to diagnose faster and more accurate. **Why Deep Learning (DL)?** Traditional machine learning relies on handcrafted features (e.g., wavelet energy), which are sensitive to noise and require expert tuning. DL, however, learns hierarchical representations directly from raw EEG, automating feature extraction and classification. The 2021 review by Shoeibi et al. confirms that DL consistently outperforms traditional methods on public EEG datasets, especially in multi-channel, long-duration recordings. **Data Collection:** Data will be sourced from publicly available anonymized EEG datasets (e.g., CHB-MIT Scalp EEG Database or Temple University Hospital EEG Corpus), ensuring no direct interaction with patients. All data is pre-collected, pre-processed, and anonymized, eliminating the need for informed consent. **Proposed Solution:** The core of my epilepsy detection system is a hybrid deep learning architecture combining 1D Convolutional Neural Networks (1D-CNN), Bidirectional Long Short-Term Memory (Bi-LSTM) layers, and a Self-Attention mechanism. This design is specifically tailored to the unique challenges of multi-channel EEG signals—high-dimensional, non-stationary time series with subtle, transient pathological patterns that traditional methods often fail to capture reliably. My understanding of EEG dynamics and deep learning principles drives this choice: seizures are not isolated events but evolve over time with localized spikes, rhythmic discharges, and channel-specific correlations, requiring a model that extracts hierarchical features while preserving temporal context and focusing on clinically meaningful segments.				

- **1D-CNN** extracts local spatio-temporal patterns (e.g., spike-wave complexes) across channels.
- **Bi-LSTM** models long-term temporal dependencies (seizure evolution over seconds).
- **Self-Attention** dynamically focuses on seizure onset windows, enhancing interpretability. This hybrid architecture is ideal for multi-channel, variable-length EEG and has shown >98% accuracy in recent cross-dataset studies.

Implementation Plan:

1.Pre-processing (MNE-Python):
- Resample all data to **256 Hz** (unifies sampling, preserves <45 Hz seizure bands).
- Apply **0.5–45 Hz bandpass filter** (removes drift, EMG, power-line noise).
- **Z-score per channel** (normalizes amplitude across patients/devices).
- Segment into **4-second windows, 50% overlap** (captures full seizure dynamics, increases data density).

2.Training:
- Pre-train on 80% of TUH (large, diverse).
- Fine-tune on CHB-MIT with domain adaptation
- Optimizer: AdamW, lr=1e-4, early stopping.

3.Baselines:
- **SVM + wavelet-packet features** (traditional ML).
- **Pure 1D-CNN** (no LSTM/Attention) — to validate sequential modeling.

4.Evaluation: 10-fold **patient-level** cross-validation; metrics: Accuracy, Sensitivity, Specificity, F1, AUC.

5.Output: Trained .h5 model, performance tables. Results will be exported as PDF reports.

Tools and Hardware:
Language: Python 3.9+
Libraries: MNE-Python, TensorFlow with Keras API, Scikit-learn, Matplotlib
Hardware: Google Colab Pro (GPU), EECS JupyterHub, local laptop for development
Dataset: Public EEG datasets (anonymized)

References:
1. Acharya UR, Oh SL, Hagiwara Y, Tan JH, Adeli H. Deep convolutional neural network for the automated detection and diagnosis of seizure using EEG signals. Comput Biol Med. 2018 Sep 1;100:270-278.
2. Acharya UR, et al. Automated EEG-based screening of depression using deep convolutional neural network. Comput Methods Programs Biomed. 2018;161:103–13.
3.Goldberger AL, et al. PhysioBank, PhysioToolkit, and PhysioNet. *Circulation*. 2000;101(23):e215–20.
4. Obeid I, Picone J. The Temple University Hospital EEG Data Corpus. *Front Neurosci*. 2016;10:196.
5. Shoeb AH, Guttag JV. Application of machine learning to epileptic seizure detection. In: Proceedings of the 27th International Conference on Machine Learning (ICML'10). 2010. p. 975–982.

	6. Shoeibi, Afshin, Marjane Khodatars, Navid Ghassemi, Mahboobeh Jafari, Parisa, Moridian, Roohallah Alizadehsani, Maryam Panahiazar, Fahime Khozeimeh, Assef Zare, Hossein Hosseini-Nejad, and et al. 2021. "Epileptic Seizures Detection Using Deep Learning Techniques: A Review" International Journal of Environmental Research and Public Health 18, no. 11: 5780.
道德规范 **Ethics**	Please confirm by checking the box: ☒ I confirm that I have discussed ethical issues with my supervisor.

Please discuss ethical issues with your supervisor. **Please refer to Project Student Handbook section 4.1 and Project Ethics Guidance**	**Summary of ethical issues:** I have discussed the ethical implications of this project with my supervisor. This project does not involve human participants or the collection of primary data from individuals. All EEG data will be obtained from publicly available, fully anonymized datasets (e.g., CHB-MIT, TUH EEG Corpus), which have been pre-approved for research use. However, as the project uses Artificial Intelligence (AI) techniques, the following ethical considerations apply: **Data Provenance:** All datasets and code will be properly cited using Vancouver referencing. No data or code will be used without attribution. **Data Confidentiality:** Datasets are fully anonymised at source. No personally identifiable information will be processed. Data will be stored securely on password-protected cloud storage (Google Drive) and deleted at the end of the project. **Addressing Bias:** Class imbalance and demographic limitations (e.g., paediatric focus in CHB-MIT) will be analysed and mitigated through stratified sampling, weighted loss function, and ablation studies. **Societal Impact:** The system is intended solely for research and diagnostic support. It will not be deployed in clinical settings without further validation. Potential misuse (e.g., over-reliance leading to misdiagnosis) will be acknowledged.
中期目标 **Mid-term target.** **It must be tangible outcomes, E.g. software, hardware or simulation.** **It will be assessed at the mid-term oral.**	**Mid-term target (March 2025)** By the mid-term assessment, the following tangible outcomes will be achieved: 1. **Dataset acquired and pre-processed**: Full EEG dataset downloaded, filtered, segmented, and split into train/validation/test sets. 2. **DLN model implemented**: Proposed DLN model built in TensorFlow/Keras with working training loop. 3. **Initial training completed**: Model trained with recorded loss/accuracy curves. 4. **Preliminary results**: Accuracy, sensitivity, and confusion matrix reported on validation set.

Work Plan (Gantt Chart)

Fill in the sub-tasks and insert a letter X in the cells to show the extent of each task

	Nov 1-15	Nov 16-30	Dec 1-15	Dec 16-31	Jan 1-15	Jan 16-31	Feb 1-15	Feb 16-28	Mar 1-15	Mar 16-31	Apr 1-15	Apr 16-30
Task 1 [To learn Deep Learning Network and EEG signal classification]												
Literature review on DLN and EEG epilepsy	X	X	X									
Study MNE-Python and EEG preprocessing	X	X	X	X								
Datasets collection and exploration		X	X	X	X							
Task 2 [To develop the DLN model for classifying Epilepsy patient EEG signals]												
Design 1D-CNN architecture (DLN)			X	X	X							
Implement preprocessing pipeline				X	X	X						
Data harmonization and cleaning					X	X	X					
Task 3 [To train the Deep Learning model with the EEG data]												
Implement 5-fold stratified cross-validation					X	X	X					
Train 1D-CNN with merged dataset						X	X	X	X			
Hyperparameter tuning and ablation study							X	X	X	X		
Task 4 [To compare the performance of DLN model with other DL models]												
Implement LSTM baseline								X	X	X		
Implement SVM with spectral features									X	X	X	
Performance comparison and statistical analysis										X	X	X